NATIONAL MATHEMATICS CONTEST

O-LEVEL PAPER 1.

SATURDAY 18TH MARCH, 2017 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are* **two** *sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*). Marks will be awarded for only answers for which a clear and logical layout of the working is provided.*

(c) *Answer scripts of only qualified students should be returned to the contest coordinator, latest* **Monday 20th March, 2017**.

(d) **ALL** *participants who qualify for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 29*th July,* 2017 *at NOON at Makerere University, Kampala.*

(e) *National Mathematics Contest* 2017 *Paper* 2 *will be done on* 17*th June,* 2017 *at various centres.*

SECTION A : 5 marks each

1. If $2x = 3y + 5$. What is the value of $\frac{4^x}{8^y}$?

2. What is the smallest positive integer which is divisible by all 1 digit primes but has no prime digits?

3. Define a function f on the set of integers by

$$f(n) = \begin{cases} n-1 & \text{if } n \text{ is even} \\ n^2-1 & \text{if } n \text{ is odd.} \end{cases}$$

   List all the values of n for which $f(f(n)) = 8$.

4. How many of the two digit numbers between 10 and 99 inclusive have the property that both are perfect squares and do not start with a 4.

5. John did push ups everyday for 7 days. Each day after the first day, he did 5 push ups more than the day before. In total he did 175 push ups. How many push ups did John do on the third day.

6. Find all positive real numbers x for which

$$\log_4 x - \log_x 16 = \frac{7}{6} - \log_x 8.$$

7. A square $ABCD$ has sides of length $3cm$. Side AB is extended through B to E with $BE = 1cm$. Segment DE intersects BC at point F. What is the area of $\triangle CDF$?

8. A Politician A lies on Mondays, Tuesdays and Wednesdays and tells the truth on other days of the week. Politician B lies on Thursdays, Fridays and Saturdays and tells the truth on other days of the week. One day both of them say " Yesterday was one of my lying days". What day is it when they say this if one says the truth and the other one lies?

9. Suppose a given population of a country increased by 20% in 2015 and decreased by 20% in 2016. What is the total percentage change over the two years period?

10. The price of a ticket was $UGX10,000$ before reduction. When the price was reduced the number of customers increased by 50% but the amount of money collected only increased by 20%. By how much was the ticket price reduced?

SECTION B : 10 **marks each**

11. Georges' car gets 3 more kilometers per litre during highway driving than it does during city driving. On a recent trip he drove 112 kilometers on the highway and 150 kilometers in the city and used exactly 10 litres of petrol. How many kilometers per litre does his car use during city driving?

12. Arrange the numbers from 1 to 9 in the 3×3 grid below so that each number occurs once, the product of the entries in the first row is 12. Product of entries in the second row is 112, product of entries in the first column is 216 and that of the second column is 12. Find the value of $A+B+C$.

A		
	B	
		C

13. If Bonny and Mary leave their houses at the same time, walking directly towards each other, each at their own constant rate, they will meet after 5 minutes. If Bonny leaves 3 minutes later than Mary they meet after Bonny has walked for 3 minutes. How many minutes would it take him to walk all the way from his house to Mary's?

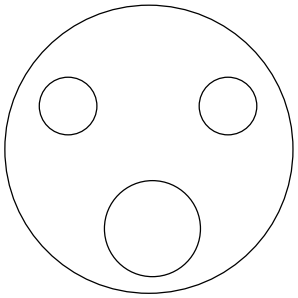
14. Find the ordered triple(s) of real numbers (a, b, c) which satisfy;

$$a^2 + 1 = 3b + c$$

$$b^2 + 33 = 7c - 3a$$

$$c^2 - 5 = b - 3a$$

15. The radius of each small circle is $\frac{1}{6}$ of the radius of the large surrounding circle. The radius of the middle sized circle is twice that of either small circle. What fraction of the area of the large circle lies outside the three interior circles?



SECTION A : 5 marks each

Qn. 1.

$$\frac{4^x}{8^y} = \frac{2^{2x}}{2^{3y}} = 2^{2x-3y} = 2^5$$

since $2x - 3y = 5$.

Qn. 2. The number must be a multiple of $2 \cdot 3 \cdot 5 \cdot 7 = 210$ that is $210, 420, 630$ or $840 \Rightarrow$ the requested number is 840.

Qn. 3. $f(n)$ is odd when n is even and even when n is odd so,

$$f(f(n)) = \begin{cases} (n-1)^2 - 1 & \text{if } n \text{ is even} \\ n^2 - 2 & \text{if } n \text{ is odd.} \end{cases}$$

Now if $f(f(n)) = 8 \Rightarrow (n-1)^2 - 1 = 8 \Rightarrow n = -2$ or $n = 4$.

Qn. 4. The digits that form two digit perfect squares are $0, 1, 4, 9$ so that there are three choices to fill the first position and four choices for the second position thus there are twelve numbers whose two digits are perfect squares.

Excluding those that begin with a digit 4 that is $40, 41, 44$ and 49 there are 8 digits that have both digits as perfect squares. $[10, 11, 14, 19, 90, 91, 94, 99]$

Qn. 5. John did 175 push ups for 7 days (an odd number of days) and on each day he did an equal number of push ups more than the day before (5 more).

The number of push ups that John did on the middle day (Day 4) is equal to the average number of push ups that he completed each day, Mean $\bar{X} = \frac{175}{7} = 25$ push ups per day. On day 4 he did 25 push ups on the third day.

DAY I - 10, DAY II - 15, DAY III - 20, DAY IV - 25, DAY V - 30, DAY VI - 35, DAY VII - 35

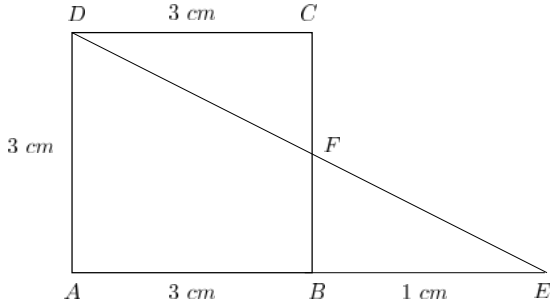
Qn. 6.

$$\begin{aligned} \log_4 x - \log_x 16 &= \frac{7}{6} - \log_x 8. \\ \log_4 x &= \frac{7}{6} + \log_x 16 - \log_x 8 \\ &= \frac{7}{6} + \log_x 2 \\ &= \frac{7}{6} + \frac{\log_4 2}{\log_4 x} \end{aligned}$$

Let $\log_4 x = y$ so that we have $y = \frac{7}{6} + \frac{\frac{1}{2}}{y}$ on solving we get $y = \frac{3}{2}$ and $y = -\frac{1}{3}$ thus

$$x = 8 \text{ or } x = \frac{1}{4^{\frac{1}{3}}}.$$

Qn. 7. From the diagram By similarity of triangles, $\frac{BF}{BE} = \frac{3}{4}$ thus $BF = \frac{3}{4}BE$. Now $CF = 3 - \frac{3}{4} = \frac{9}{4}$.



The area

$$A = \frac{1}{2}b \cdot h = \frac{1}{2} \cdot 3 \cdot \frac{9}{4} = \frac{27}{8}cm^2.$$

Qn. 8. The only days that politician A would make that statement are Monday and Thursday. The only days that politician B would make that statement are Thursday and Sunday $\Rightarrow$ both made the statement on Thursday if one lies and the other says the truth.

Qn. 9. Let the population of the country be N before increase then N would increase to $1.2N$ and then a decrease of 20% would lead to

$$0.8 \cdot (1.2 \cdot N) = 0.96N.$$

Therefore the total change is $N - 0.96N = 0.04N$ equivalent to 4%.

Qn. 10. If the new fee is x and the original number of customers was N then

$$1.2 \times 10,000 \cdot N = 1.5Nx \;\Rightarrow\; x = \frac{12000}{1.5} = 8,000/=$$

so the ticket was reduced by $2,000/=$.

SECTION B : 10 marks each

Qn. 11. Let x denote the kilometers his car gets during city driving then

$$\frac{112}{x+3}+\frac{150}{x}=10 \Rightarrow (5x+9)(x-25)=0 \Rightarrow$$

either $x=-\frac{9}{5}$ or $x=25$. Therefore the number of kilometers his car gets during city driving is 25km/litre.

Qn. 12. We can fill in the $7, 5$ and then the 4 since the last column needs 7 and 5 powers of 2. We can fill in 8 and 2 in the second row then the 6 and 1 in the second column and finally the 3 and 9 in the first column.

3	1	4	$=2^2\cdot 3$
8	2	7	$=2^4\cdot 7$
9	6	5	
$2^3\cdot 3^3$	$2^2\cdot 3$		

$A+B+C=3+2+5=10$.

Qn. 13. Let S_B and S_M denote the speeds of Bonny and Mary respectively. Let d be the distance between their houses, then

$$5S_B+5S_M=d=3S_B+6S_M \;\Rightarrow\; S_M=2S_B$$

and

$$d=15S_B$$

therefore Bonny takes 15 minutes to walk all the way to Mary's house.

Qn. 14. Add all the equations to get;

$$a^2+b^2+c^2+29=4b+8c-6a$$

Combining common terms we get

$$(a^2+6a)+(b^2-4b)+(c^2-8c)=-29$$

Completing squares we have

$$(a+3)^2+(b-2)^2+(c-4)^2=0 \;\Rightarrow\; a=-3,\ b=2 \text{ and } \ c=4$$

Qn. 15. Let the radius of the large circle be 1. The sum of the areas of three circles is

$$\pi\left(\frac{1}{36}+\frac{1}{36}+\frac{1}{9}\right)=\frac{\pi}{6}$$

therefore the fraction is $\frac{5}{6}$ of the total area π.